AGUAS ZARCAS (CM2) LOW TEMPERATURE THERMOPHYSICAL PROPERTIES

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Abstract: **Measurement of the low-temperature thermodynamic and physical properties of meteorites provides fundamental data for the study and understanding of asteroids and other small bodies. Of particular interest are the CM carbonaceous chondrites, which represent a class of primitive meteorites that record substantial chemical information concerning the evolution of the early solar system. Aguas Zarcas (AZ) is a brecciated carbonaceous chondrite (CM2) containing both chondrule-poor and chondrule-rich lithologies [1]. The AZ samples collected immediately after the fall (on 23 April 2019 in Aguas Zarcas, Costa Rica) provide a rare opportunity to analyze the mineralogy of a CM2 largely free of terrestrial contamination [2,3,4]. AZ contains both anhydrous minerals such as olivine and pyroxene found in chondrules, along with abundant hydrous phyllosilicates in the meteorite matrix, interspersed between the chondrules.**

We report low-temperature specific heat capacity, thermal conductivity, and linear thermal expansion data over the range $5 \leq T \leq 300$ K. Bulk density was obtained by the Archimedes method after completing the thermodynamic measurements. From these data, we calculate thermal inertia and thermal diffusivity as a function of temperature. Measurements follow the procedures as described in [5] utilizing a Quantum Design Physical Property Measurement System (QD-PPMS) at Boston College. Our primary reason for presenting these results are the petrological comparisons noted [1,3,6,7,8] between AZ and the samples returned from the asteroid 101955 Bennu by the OSIRIS-REx mission. Recent thermal diffusivity measurements on a Bennu sample record a value of 3.80 x 10^{-7} (m^2 s^{-1}) at 300 K; this compares with our AZ value of 2.22 x 10^{-7} (m^2 s^{-1}) at 300 K. Similar thermophysical measurements are planned in the Opeil Lab on the Bennu samples as they become available.

(1) Introduction:

(a) Heat capacity, thermal conductivity, thermal diffusivity, thermal inertia, bulk density, porosity and thermal expansion are the fundamental thermodynamic and physical meteorite characteristics. Such data help to model various aspects of asteroidal surface properties, as well as, the internal evolution of asteroids and small solar system bodies. These characterization techniques help to identify material and composition dynamics of the meteorite itself.

(b) Kerraouch et al. [1] argue that Aguas Zarcas (CM2) is petrologically similar to samples of 101955 Bennu retrieved by OSIRIS REx. Bennu's samples are likely thought to contain phyllosilicates, a type of clay, resulting from aqueous alteration of the parent body minerals. [1,2,3,5,6,8,11] Other CM2 meteorites studied [5] exhibit similar thermophysical characteristics, shown in Fig. 3, particularly the negative thermal expansion (NTE). The unusually large NTE at ~235 K may be responsible for rock fractures on asteroidal surfaces due to uneven surface heating/cooling over time.

(c) The negative thermal expansion shown in Fig. 3 is thought to be the result of transverse vibrations of molecules that link the layers of phyllosilicates present in the meteorite. The transverse vibrations contract the phyllosilicate layers and are most energetic in the temperature range of 200-240 K. [5] Coincidently, the temperature on the surface of asteroid Bennu fluctuates between 200 and 390 K; the T depends on asteroid surface latitude and distance from the Sun.

(2) Specific Heat Capacity (c_p) and Thermal Conductivity (κ) vs. T (5-300 K)

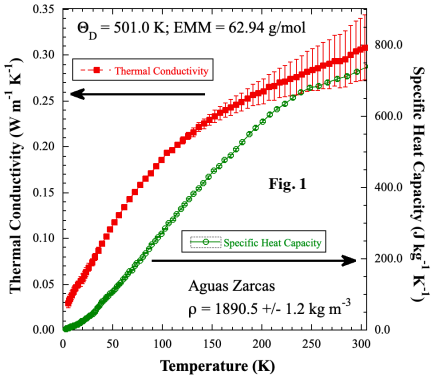

Fig. 1: Specific heat capacity vs. T data (right axis, open circles) and thermal conductivity vs. T data (left axis, filled squares) are shown. Debye temperature, Θ_D, and effective molar mass (EMM) calculated by numerical methods from specific heat capacity data.

(3) Thermal Diffusivity (D_T) and Thermal Inertia (Γ) vs. T (5-300 K)

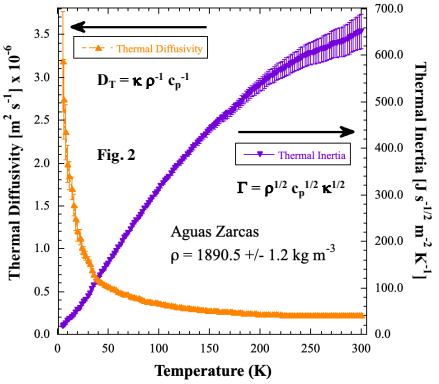

Fig. 2: Calculated thermal inertia vs. T data (right axis, unfilled inverted triangles) and thermal diffusivity vs. T data (left axis, filled triangles) are shown. Bulk density, ρ, measured by Archimedes method.

(4) Thermal Expansion Coefficient (α) vs. T

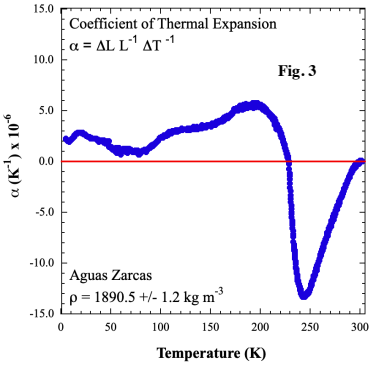

Fig. 3: Thermal expansion coefficient (α) vs. T measurements are shown. Data were taken at a cooling rate of 0.3 K/min using a Schmiedeshoff capacitive dilatometer [13] with a resolution of ~ 0.5 x 10^{-10} m.

(5) Aguas Zarcas (CM2) and 101955 Bennu Asteroid Sample Photos

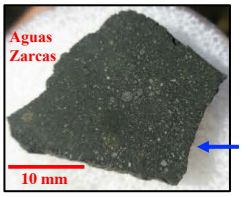

Photo 1: Aguas Zarcas (CM2) carbonaceous chondrite sample slice, m = 1.16 g. Meteorite was an observed 'fall' on 23 April 2019. The sample (at left) was used for thermophysical data measurements shown above. (Opeil photo, 30 June 2019)

Photo 2: Asteroid 101955 Bennu samples shown in the TAGSAM module of the NASA Mission OSIRIS REx. Samples were retrieved from Bennu on 10 October 2020 & landed in Utah Test & Training Range on 19 September 2023. These samples travelled ~ 1.8 billion miles & are part of ~ 70.0 g capture of material. (NASA photo, 17 Jan 2024)

(6) Conclusions/Analysis:

(a) A range of experiments on the Bennu samples are currently underway. Specific heat capacity and thermal expansion coefficient measurements of the Bennu are planned for the Opeil Lab.

(b) If the predictions of [1,3,6,7,8] regarding the Aguas Zarcas & Bennu petrology are correct, no one should be surprised to discover similar negative thermal expansion (NTE) and specific heat behaviors in the Bennu samples as revealed for Aguas Zarcas in Figs. 1 and 3.

(c) OSIRIS REx found 101905 Bennu to be a ruble pile asteroid with minimal gravity. If NTE behaviors are indicated for Bennu, there is a likely correlation between the surface rock fracturing and this NTE behavior. [9,10,11,12]

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References: [1] Kerraouch I. et al. (2021) *Meteoritics & Planet. Sci.,* 56:277-310. [2] Tunney L. D. et al. (2022) *Meteoritics & Planet. Sci.,* 57:883-911. [3] Garvie L. A. J. (2021) *Amer. Miner.* 106:1900-1916. [4] Gattacceca, J. et al. (2020) *Met. Bull. 108. Meteoritics & Planet. Sci.* 55:1146–1150. [5] Opeil C. et al. (2020) *Meteoritics & Planet. Sci.,* 55: E1-E20. [6] Clark B. E. et al. (2023) *Icarus* 400: 115563. [7] DellaGiustina D. N. (2019) *Nature Astronomy* 3:341-351. [8] Dionnet Z. et al. (2022) *Meteoritics & Planet. Sci.,* 57, 965-988. [9] Delbo M. et al. (2007) *Icarus* 190:236-249. [10] Delbo M. et al. (2014) *Nature* 508:233-236. [11] Lauretta D. S. et al. (2019) *Nature* 568:55-60. [12] Delbo, M. et al. (2022) *Nature Geoscience* 15:453-457. [13] Schmiedeshoff G. M. et al. (2006) *Rev. Sci. Instrum.* 77:123907.

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